

Kumar Anubhav

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EDUCATION

Indian Institute of Information Technology, Ranchi

Bachelor of Technology in Computer Science and Engineering (CGPA - 8.51)

2023 – 2027

Ranchi, Jharkhand

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, C, C++

Frontend & Backend: React.js, Next.js, TailwindCSS, HTML, CSS, FastAPI, Django, Flask, Express.js, Firebase

AI and ML: LangChain, FAISS, Sentence Transformers, ONNX, Groq APIs

Cloud & DevOps: Google Cloud Platform, Serverless Architecture, Render, Linux, Git, Docker

Databases: PostgreSQL, MongoDB, Firestore, Cloud SQL, Redis

Payments & APIs: Stripe API, Razorpay API, RESTful APIs, Webhooks

Coursework: Data Structures & Algorithms, Database Management Systems, Computer Networks, OOPs

EXPERIENCE

Full Stack Development Intern

Jul 2025 – Jun 2026

DAaranya.ai (Early-Stage Startup)

Remote

- Designed **UML diagrams** with the engineering manager to plan system architecture and data models before implementation across the IntelliCoach and senior-care platforms.
- Designed **event-driven processing** with **Google Cloud Pub/Sub** for asynchronous bulk-user provisioning and AI question-generation jobs, building **push-based** (Cloud Run/FastAPI HTTP endpoints) workers with idempotent **ACK/NACK** redelivery, per-user savepoints, and **backoff/jitter** retries for Gemini rate-limits.
- Developed **multi-tenant JWT authentication** (short-lived access / refresh tokens) and organization/facility-scoped **RBAC** for a Next.js/TypeScript senior-care discovery platform, including facility-claiming and manager-invitation workflows.
- Contributed across a multi-frontend **React/Vite** architecture using **TanStack Query**, protected routing, and a **Pyodide Web Worker** to run browser-side Python without blocking the UI thread, alongside centralized error handling and structured logging in production.
- Configured asynchronous **PostgreSQL connection pools** (pre-ping, overflow, timeouts, recycling) with **PgBouncer** at the deployment layer for cross-instance connection multiplexing, plus centralized validation, structured errors, and sensitive-field redaction.
- Controlled concurrent **Gemini vision** calls with **asyncio.Semaphore** and request timeouts, and worked with course/module/lesson domain models, progress tracking, and gamification structures for the IntelliCoach platform.

PROJECTS

Lumina | Agentic Academic Operating System for ADHD-Friendly Planning [GitHub](#)

Jul 2026 – Present

- Stack:** Python, FastAPI, SQLAlchemy, Neon Postgres, Telegram, LangChain, Groq, Upstash Redis
- Architected and deployed a Telegram-based academic assistant for students with **ADHD**, combining **deterministic attendance-risk analysis**, timetable-aware goal scheduling, exam tracking, syllabus monitoring, and proactive reminders.
- Engineered pure Python attendance and scheduling engines using **exact rational arithmetic**, free-gap detection, and minimum-viable-effort negotiation to produce explainable plans **without LLM-generated calculations**.
- Built a **multimodal timetable-ingestion pipeline** (Groq vision extraction, preview-based confirmation) within an **AI-at-the-edge architecture** where LangChain/Groq models only explain or recommend actions, backed by a durable **Upstash Redis retry queue** for reliable delivery.

PulseRAG | Audio Conversational Intelligence and RAG System [GitHub](#)

Jun 2026 – Present

- Stack:** Python, FastAPI, FAISS, Groq Whisper, Llama 3.3, Mistral Embeddings, Redis, Docker, Render
- Built an end-to-end **audio-to-RAG pipeline** using FastAPI, Groq Whisper, FAISS, and LLMs to transcribe conversations, label speakers, index transcript chunks, and answer questions with **source-grounded responses** via asynchronous ingestion (**25 MB** limit).
- Reduced application boot memory from approximately **755 MB to 80 MB** via a **pluggable embedding layer** and **Redis response caching**; validated with **72 automated tests** and containerized via Docker/Render.

ACHIEVEMENTS AND LEADERSHIP

- Organizing team member for Quasar 2024 Hackathon and Competitive Programming Contest at IIIT Ranchi.
- Lead, IoT and Robotics Club, IIIT Ranchi.